Sankalp Jha

Portfolio | LinkedIn | GitHub

Open Source Developer | Systems Engineer | Technical Content Writer | Public Speaker

sankalp.jha9643@gmail.com

Delhi NCR, India

+91-8800117422

Education

Year	Degree/Certificate	Institute	CPI/%
Oct 2022 – Aug 2026	B.Tech in CS & IT	Ajay Kumar Garg Engineering College, Delhi NCR	7.25/10
2021 – 2022	Senior Secondary (XII)	New Era School, Delhi NCR	91.2%
2017 – 2020	High School (X)	SJN International School, Coimbatore	92.6%

Experience

Google Summer of Code 2025 Contributor | *P4 Language Consortium, Stanford, CA* (*May 2025 – Sept 2025*)

Project: SpliDT: Scaling Stateful Decision Tree Algorithms in P4 *P4, Python, P4Runtime, Docker, C++, Intel Tofino*

- **Switch-Native Architecture:** Architected a framework to compile ML decision trees directly into P4 data planes, implementing Partitioned Decision Trees (PDTs) with TCAM-efficient Subtree IDs (SIDs) to resolve hardware ASIC constraints.
- **Automated Validation:** Engineered a Jinja2-based testing pipeline automating 25+ P4Runtime validations across Intel Tofino and Mininet environments, improving deployment reliability by 30%.
- **Performance Optimization:** Achieved 10 Gbps line-rate throughput and reduced packet inference latency by 20% (5x lower than control-plane inference) via optimized SID recirculation.

Open Source Contributions

- **WasmEdge/WasmEdge:** #4470 fix(wasi): resolve 100% CPU usage on macOS due to clock mismatch; #4440 Upgrade Alpine base to 3.23 and restore static LLD support; #4466 feat(docker): add standalone alpine-base image build pipeline; #4452 docs(README): fix broken introduction and blog links.
- **p4lang/tutorials:** #681 Update README.md in reference to #680.
- **facebook/bpfilter:** #507 style: add missing braces to satisfy readability-braces-around-statem....

Key Projects

Los-Tecnicos | *Go, TypeScript, Rust, Soroban, Raspberry Pi, ESP32* (*Project Link*)

Objective: Build a decentralized energy grid that tokenizes renewable-energy production on the Stellar ecosystem.

Approach: • Built a Go backend and three Soroban smart contracts. • Integrated a Raspberry Pi and ESP32 hardware mesh with the software platform.

Impact: Won Asia-Pacific Web3 bounties and received recognition from the Government of India.

DMIE | *Next.js, TypeScript, Python, FastAPI, Vector DB, LLM* (*Project Link*)

Objective: Determine whether a startup idea already exists by comparing it against more than 500,000 company records.

Approach: • Built a scheduled scraper that keeps the startup dataset current. • Combined LLM query understanding with vector search and backend filters.

Impact: Raised USD 10,000 and serves more than 100 users worldwide.

Do-It | *Go, SSE, HTTP, Raspberry Pi, Local-first* (*Project Link*)

Objective: Provide private, synchronized notes and media across devices on a local network.

Approach: • Packaged a lightweight Go SSE and HTTP server for Raspberry Pi, routers, and repurposed phones.

Impact: Delivers self-hosted, account-free synchronization without sending household data to a cloud service.

GSoC- SpliDT: Scaling Stateful Decision Tree Algorithms in P4 | *P4, Python, P4Runtime, Docker, C++, Intel Tofino* (*Project Link*)

Objective: Compile machine-learning decision trees directly into P4 data planes using Partitioned Decision Trees.

Approach: • Architected a switch-native inference framework around Subtree ID recirculation. • Automated P4Runtime validation across Tofino and Mininet test environments.

Impact: Achieved 10 Gbps line-rate throughput and 5x lower latency than control-plane inference.

p4Lens | *React, FastAPI, Docker, React Flow, P4* (*Project Link*)

Objective: Make P4 programs easier to learn and debug through an interactive visualization of their processing pipeline.

Approach: • Built a parser for P4-14 and P4-16 programs. • Rendered headers, parsers, controls, and deparsers as interactive React Flow topologies.

Impact: Renders pipeline diagrams in under 200 ms and reduces control-plane debugging effort by approximately 40 percent.

Slix | *React, Go, Azure OpenAI, REST API* (*Project Link*)

Objective: Combine movie streaming, reviews, and machine-generated sentiment summaries in one full-stack application.

Approach: • Built the frontend in React and the service layer in Go. • Added Azure OpenAI sentiment analysis for curator reviews.

Impact: Gives viewers a concise representation of a movie's tone before watching.

Moodish | *JavaScript, Node.js, OpenAI API, MCP* (*Project Link*)

Objective: Rank meal options from a user's craving, budget, and dietary constraints while keeping the recommendation explainable.

Approach: • Built intent extraction and deterministic ranking around an AI-generated summary. • Added a recommendation trace and confirmation-gated cart preview.

Impact: Produces safer, inspectable food recommendations instead of opaque model output.

ebpf-mcp-tracer | *Python, bpftrace, eBPF, Linux, MCP* (*Project Link*)

Objective: Let AI debugging agents inspect Linux performance without granting unrestricted eBPF execution.

Approach: • Implemented a strict probe allowlist, dry-run validation, and execution timeouts around bpftrace. • Exposed the guarded tracing workflow through MCP.

Impact: Enables kernel-level diagnostics while reducing command-injection and runaway-probe risk.

Myprod | *Go, Nomad, Traefik, WireGuard, Docker, Netlify DNS, SOPS* (*Project Link*)

Objective: Turn manually provisioned VPS machines into a low-overhead personal compute pool for backend services.

Approach: • Built a Go CLI and operator dashboard around Nomad scheduling, Traefik ingress, and WireGuard networking. • Added guarded node lifecycle controls, application registration, live resource metrics, and idempotent Netlify DNS management.

Impact: Provides one control surface for deploying applications and managing heterogeneous free-tier and temporary compute capacity.

Open Claw Lab | *Node.js, OpenClaw, Gemini, Oracle Cloud, systemd, Telegram* (*Project Link*)

Objective: Discover high-signal engineering opportunities while preventing unsupported or stale leads from reaching the user.

Approach: • Built a multi-agent discovery and independent-review pipeline around typed lead records. • Added deterministic freshness, eligibility, contact relevance, relocation evidence, and deduplication gates.

Impact: Delivers an evidence-backed Telegram digest while rejecting guessed contacts, unsupported geography, stale roles, and duplicate leads.

Technical Skills

- Python, C, C++, JavaScript, TypeScript, Go, P4
- **Libraries/Frameworks:** NodeJS, Numpy, Matplotlib, eBPF, gRPC, Protobufs, Next.js, Scapy, MLIR, LLVM, Sklearn, React, Web Services
- **Tools/Platforms:** Docker, CI/CD, VS Code, UTM, GitHub, GitLab, DaVinci Resolve, Linux (x86_64, ARM64 & AMD64)
- **Databases:** SQL, MongoDB, GraphQL, Redis, PostgreSQL, MySQL

Positions of Leadership

- **Technical Writer, Cloud Native Community Group, New Delhi** (*May 2024 – Dec 2024*)
  - Led a team of **6+** content writers and **4** designers to deliver documentation and event management support.
  - Partnered with marketing to revamp social media presence across LinkedIn, Instagram, and Twitter, achieving a **60%** increase in follower engagement.
- **Co-Organizer & Anchor, Google Developer Groups & GDSC WOW 2024, Delhi NCR** (*Nov 2023 – Nov 2025*)
  - Co-led **10+** tech events and anchored the GDSC WOW flagship event at IIIT Delhi, coordinating **31+** crew members across web, social, and on-ground operations.
  - Engaged a live audience of **800+** at WOW and **500+** developers overall, boosting participation by **30%** and visibility by **40%**.

Publications

- R. Bharadwaj, **S. Jha**, V. Malyan, R. Gupta, "Trusture: A Blockchain-Based Decentralized Framework for Secure, Transparent, and Auditable NGO Transactions," *2026 International Conference on Computing, Sciences and Communications (ICCSC)*, Ghaziabad, India, Feb. 2026. IEEE Xplore ↗ DOI: 10.1109/ICCSC67078.2026.11468597

Certifications & Honors

Certifications	• **Open Science 101** – NASA • **Data Science for Engineers** – NPTEL • **Enhancing Soft Skills & Personality** – NPTEL • **IIRS Satellite Remote Sensing** – ISRO
Honors & Awards	• **2x** AKTU State Men Singles TT Gold (2023, 2024) • International Robotics Olympiad Bronze, Thailand (2019) • Merit Certification at Carnic Model United Nation (2021) • NCC Cadet, National Cadet Corps (2019)